Ifeoluwanimi Praise Shobayo

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PROFESSIONAL SUMMARY

MSPH Epidemiology student (4.0 GPA) with applied biostatistics training and hands-on analysis in SAS, R, and Stata. Experience includes linear mixed-effects modelling of wearable health data, weighted prevalence and trend analysis of CDC PRAMS, regression and chi-square analysis, data cleaning and quality checks, and clear reporting of results and limitations. Biostatistics tutor and workshop leader who also brings clinical-laboratory and infectious-disease research exposure.

EDUCATION

Master of Science in Public Health (MSPH), Concentration: Epidemiology Anticipated May 2027
Liberty University, Lynchburg, VA

- GPA: 4.0. **Relevant coursework:** Biostatistics, Statistical Software (SAS), Epidemiologic Research Methods, Advanced Research Methods in Epidemiology, Research Methods, Data Analysis, Data Visualization & Spatial Analysis.

Bachelor of Science in Biological Sciences Mar 2023
Federal University of Agriculture, Abeokuta, Nigeria

PROFESSIONAL AND RESEARCH EXPERIENCE

Biostatistics and Research Assistant Jan 2026 – Aug 2026
Liberty University, Department of Public Health, Lynchburg, VA

- Conducted independent secondary analyses of wearable-derived health data (mcPHASES, PhysioNet; 42 participants, 4,893 person-days) using linear mixed-effects models to account for repeated measures within participants.
- Merged and cleaned multi-file person-day records, ran quality checks, and documented exclusions and the analytic sample.
- Supported faculty with data preparation, statistical analysis, and interpretation for public health and epidemiologic research projects.
- Tutored residential and online students in biostatistics; created Excel-based tutorials and instructional materials.
- Developed and facilitated biostatistics and research workshops to improve statistical literacy and applied data analysis.

Graduate Student Assistant Aug 2026 – Present
Liberty University, Department of Public Health, Lynchburg, VA

- Teach class sessions, facilitate discussions, activities, and student presentations, and grade assignments in undergraduate public health courses.
- Assist faculty with preparing grant applications through Liberty's Office of Sponsored Programs & Research (OSPR).

Research Assistant Jul 2024 – Nov 2024
University College Hospital (UCH), College of Medicine, Department of Public Health, Ibadan, Nigeria

- Conducted health surveys with 100 workers across police stations in Ibadan for an occupational health study on workplace safety and disease exposure, using Kobo Collect for mobile data collection.
- Ensured data accuracy and confidentiality, improving research quality and processing time.
- Prepared and presented study reports and supported faculty with data visualization and communication of findings.

Research and Laboratory Assistant Apr 2021 – Oct 2021
Institute for Advanced Medical Research & Training (IAMRAT), UCH, Ibadan, Nigeria

- Six-laboratory rotational placement (malaria, genetics, virology, cancer, neurobiobanking, COVID-19 response) with data entry, record management, and quality-controlled documentation.

STATISTICAL ANALYSIS PROJECTS

- **Menstrual cycle, stress, and sleep in wearable data** (analyst; mcPHASES, PhysioNet): linear mixed-effects models on 3,732 person-days from 36 people; fertility-phase days showed more physical signs of stress than follicular days (β −0.90, 95% CI −1.43 to −0.37), the most stable finding in sensitivity analysis.
- **Perinatal depression among Virginia women, 2016–2022** (first author; CDC PRAMS; VPHA 2026): weighted prevalence with 95% CIs; self-reported depression during pregnancy rose from 9.4% to 16.9% (+79.8%) versus +44.8% across aggregate PRAMS sites.
- **Mental health and provider availability across Virginia counties** (first author; 2025 County Health Rankings, Tableau; VPHA 2026): choropleth maps and trend analyses; frequent mental distress ranged 13.5–23.6%, and childhood poverty explained far more variation ($R^2 = 0.51$) than provider supply ($R^2 = 0.08$).
- **Framingham Heart Study secondary analysis** (4,434 participants; regression, Excel): the diabetes–heart rate association (+4.3 beats/min) shrank to +0.9 and lost significance after adjustment.
- **E-cigarette use and depression among Virginia adults** (co-author; 2023 BRFSS cross-sectional analysis; APHA 2025).
- **Predicting campaign responders** (group project; Python, SQL; 11,162 records): K-Means segmentation and four classifiers with cross-validation; removed a leaking variable from the model.
- Statistical programming: SAS assumption checks, regression, chi-square relative risks, and power calculations; R tidyverse data-cleaning script; Python pandas data-quality diagnosis.

PUBLICATIONS AND PRESENTATIONS (SELECTED)

Peer-reviewed publications (co-author):

- Edeh CG, et al., **Shobayo IP**. A machine learning-based framework for real-time environmental health risk prediction and spatial air quality intelligence in urban-industrial ecosystems. *Asian J Adv Res Rep*. 2026;20(4):1-20.
- Linus EN, et al., **Shobayo IP**. Vaccine hesitancy and immunization inequity in Nigeria: public health determinants, misinformation dynamics, and health system strategies for improved uptake. *Int J Health Pharm Res*. 2026;11(8):32-56.
- Odekunle YO, **Shobayo IP**, et al. Modelling and mitigating antimicrobial resistance (AMR) through data-driven surveillance, AI-powered drug discovery, and public health intervention design. *Middle East J Appl Innov Multidiscip Res*. 2025;9(2).

Conference presentations:

- **Shobayo IP**, et al. Perinatal depression among Virginia women, 2016–2022 (CDC PRAMS); Geographic disparities in mental health across Virginia counties, 2025. VPHA 2026 Annual Conference; first author.
- Agbotui B, **Shobayo I**, Anderson R. E-cigarette use and depression among adults in Virginia: 2023 BRFSS. APHA 2025; co-author.
- Six first-author posters in total, including three accepted at APHA 2026.

AFFILIATIONS AND SERVICE

- Member, American Public Health Association (APHA) and Virginia Public Health Association. **Abstract Reviewer, APHA 2026 Annual Meeting** (Epidemiology; Epidemiology Late Breakers; Occupational Health and Safety Late Breakers; Community Health Planning and Policy Development Late Breakers).

SKILLS

Statistical methods: Linear, logistic, and mixed-effects regression; t-tests, chi-square, relative risks, confounding adjustment; power and sample-size calculations; weighted survey estimates; cross-validation
Software: SAS, R (tidyverse), Stata, Python (pandas, scikit-learn), SQL, SPSS, Excel, Tableau, Power BI
Data and reporting: Dataset preparation, data cleaning and quality control, documentation of exclusions, figures and tables, posters and manuscripts
Training and certifications: Aspire Leaders Program (2026); Data Science Program, TS Academy (2026); CyberGirls Fellowship, Cloud Security (2024); Ethical, Legal, and Social Implications of Stroke Genomics and Neurobiobanking (2023)